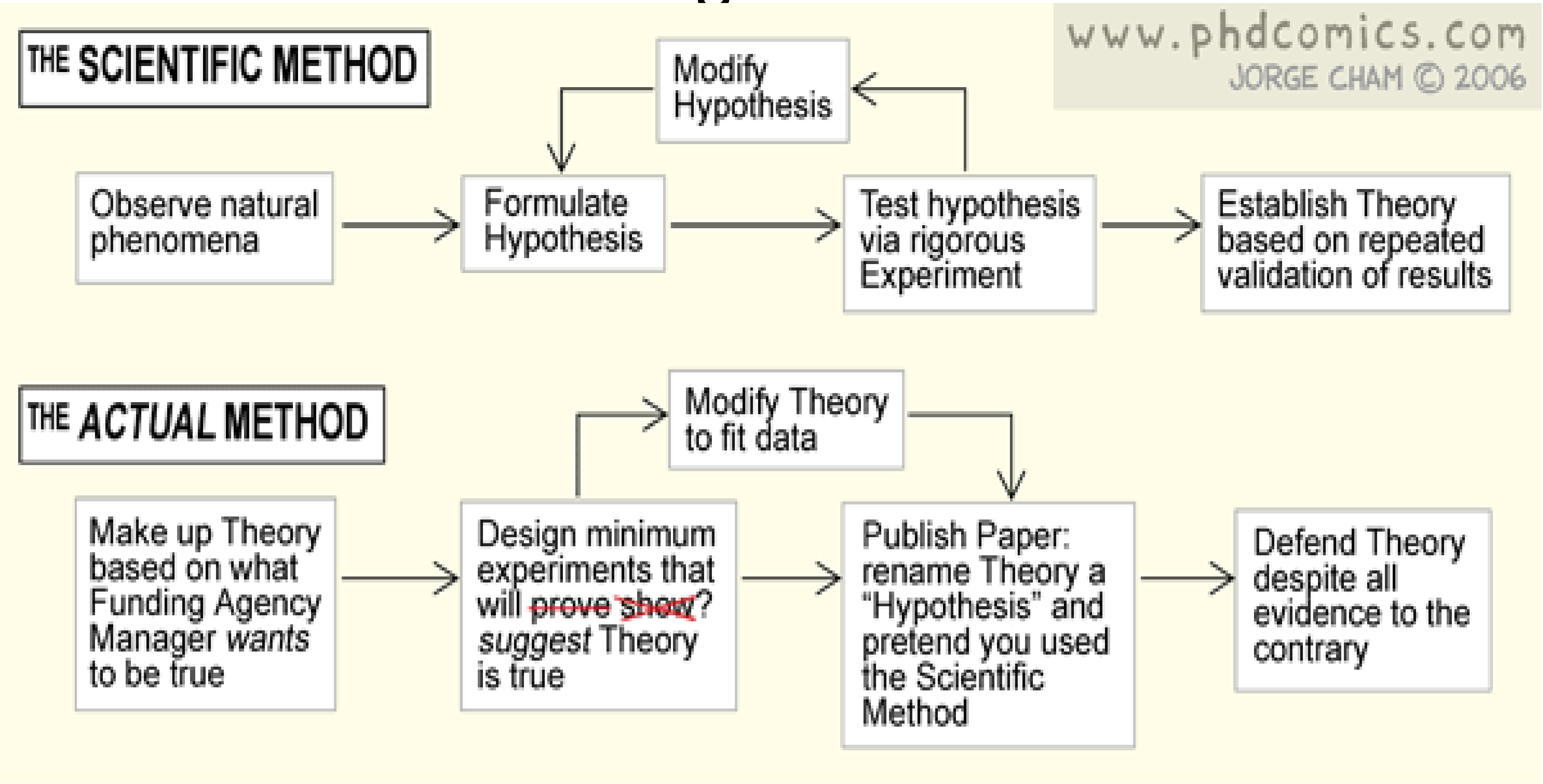
Universität Basel

Swiss TPH
Swiss Tropical and Public Health Institute
Schweizerisches Tropen- und Public Health-Institut
Institut Tropical et de Santé Publique Suisse

Associated Institute of the University of Basel

Lesson 5: Writing the Methods section
Titles and acknowledgements



Objectives Lesson 5

1. Focus on and formulate titles for thesis/article
2. Examine some acknowledgements in articles and theses
3. Examine some Methods sections:

- Explore the contents and structure of the sections
- Focus on language use in the Methods section, especially the use of tenses and the passive voice
- Practise some language structures found in this section
- Learn how to write cohesively and coherently

Feedback on assignment 3

- **Some perfect summary paraphrases!** See also the document 'Examples of good paraphrases' on ADAM.
- **Paraphrasing and summarizing are crucial skills that require much practice!**
- **Comma splice mistakes**: Use a full-stop or semi-colon – not a comma - to separate two sentences, else the result is a **comma splice**, which is a style error. Example: '~~Many children die of diarrhoea, rotavirus vaccination could reduce the number of deaths.~~' (comma splice) could be improved in several ways , for example: 'Many children die of diarrhoea; rotavirus vaccination could reduce the number of deaths.'
- **Run-on sentences:** when two sentences are joined without any punctuation marks.
- Set the **proofing language** in WORD to **English** when you write so that some grammar and spelling mistakes are automatically detected.
- Here and there inappropriate synonyms
- Practise summarizing – reflecting the gist of the text in fewer words.

Title and abstracts

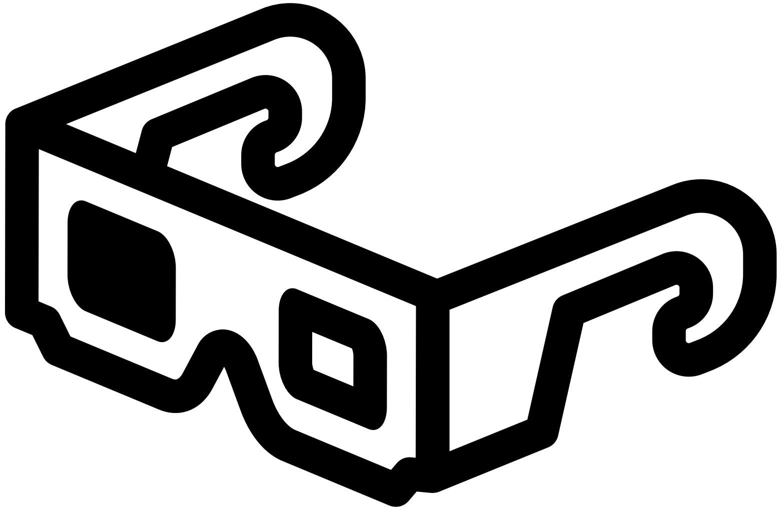
Why are titles and abstracts important? List at least three reasons.

What do you think are the requirements for a good title?

How long should a title be?

Title and abstract are important because they…

- Give readers a quick overview of your study
- Help readers decide if they want to read the rest of the text
- Form part of the indexing and compilation of reference works for printed and computerized databases, so should accurately reflect the contents of your study and include the necessary **key words** that would make them easy to find in a database



Title task 1: examine titles

- Compare titles of articles/theses with your group members. What different kinds/formats can you identify?
- Do you think the kind of article/thesis title used is field-specific (linked to the field of study)? If so, how?
- What are some requirements for a good thesis/article title?
- What is the current title of your thesis/article?
- How has the title of your thesis / article changed since you started working it?

Writing a good title: Requirements

A good title should…

- clearly indicate the topic of the study, using descriptive terms.
- indicate the scope of the study, neither overstating nor understating its significance.
- be self-explanatory to readers in that field of study.
- accurately highlight the core content of the paper (e.g., the species studied, the model tested, etc.)
- often include the nature/type of the study, for example ‘Review’ or ‘Case study’
- be about 10-12 words long – but that depends on the field of study and whether it is understandable.

Title task 2: Examine 5 titles in your field of study or refer to the list of titles in the script

1. What are the requirements for titles in your department/at the Swiss TPH/in your field of study?
2. How long are the titles in the articles you have with you?
3. Do they contain verbs? If so, which?
4. How are nouns used?
5. Do the titles contain colons (:)? If so, is the section before or after the colon longer? What do the two sections contain?
6. Do some titles contain finite (full) verbs?
7. Are there any titles that contain direct questions?
8. How are capital letters used? Most words/just first word and proper nouns?
9. Compare the titles that you found with those of your colleagues. Similarities? Differences?
10. Can you see any trends in the structure of the titles?

Consider the examples in the script too.

Various common basic title formats

Shorter main title: a longer subtitle	
Before the colon:	After the colon
Problem:	Solution
General:	Specific
Topic:	Method
Major:	Minor
Gerund form (verb+ing) (e.g., Estimating)	
Full sentences	
Questions	
Noun phrases	

Examples of titles with the colon (:) format and gerund forms (-ing) (See script for more examples)

Colon format

1. Air pollution-related illness: effects of particles (Epidemiology and public health)
2. Stunting and severe stunting among children under 5 years in Nigeria: a multilevel analysis (paediatrics, public health)
3. Genetic diversity and malaria vaccine design, testing and efficacy: preventing and overcoming ‘vaccine resistant malaria’ (Epidemiology, parasite immunology
4. Social norms and vaccine uptake: College students’ COVID vaccination intentions, attitudes and estimated peer norms and comparisons with influenza vaccine (epidemiology, preventive health, public health)

Gerund forms (-ing)

5. Treating acute diarrhea in children (health sciences)

6. Evaluating the effects of radiation from cell towers and high-tension power lines on inhabitants of buildings in Ots, Ogun State (Applied science, public health)

Sentences and questions as research article titles - examples

Full sentence containing main message/conclusion or main finding:

- Petermann ice shelf may not recover after a future breakup. (Earth and environmental sciences)
- Serum metabolome is associated with severity of acute traumatic brain injury. (Biological sciences)

Questions: Rarely used, but do occur, e.g.

- The promise of a malaria vaccine – are we closer? (microbiology, epidemiology)
- Why is Paragonite so rare in medium-temperature high-pressure rocks? (geology)

Noun phrases as titles

This format seems to be prevalent in epidemiology and public health

- Air pollution and children's health (paediatrics, public health, epidemiology)
- Novel strategies for malaria vaccine design (immunology, epidemiology)
- Impact of mobile phone radiation on different life forms (Applied science, public health)
- Epidemiology and clinical characteristics of COVID-19

Scientific writing Swiss TPH Riana Paola

As a beginner researcher, keep your titles informative rather than challenging

Well-known and respected researchers have more freedom to play around with titles than less known ones!

Stick to common basic formats (see previous slides) that include the necessary key words.

Once you are famous, you have more possibility to experiment with titles too!

Example:

An In-Depth Analysis of a Piece of Shit: Distribution of Schistosoma mansoni and Hookworm Eggs in Human Stool

Stefanie J. Krauth1,2,3, Jean T. Coulibaly1,2,3,4, Stefanie Knopp1,2, Mahamadou Traore´ 3 , Elie´zer K. N'Goran3,4, Jürg Utzinger1,2*

https://doi.org/10.1371/journal.pntd.0001969

Acknowledgements: Task

- Read acknowledgments included by the writers.
- Where are the acknowledgements printed?
- What do the writers do in the acknowledgements? That is, why do they acknowledge these people / companies?
- Does the writer use the first person (I/we)?

Acknowledgements
Examples in the script

Given just after title page of thesis, or after Conclusion of a research article

1. Financial support / Donations of materials
2. Thanks (mostly in thesis)
3. Disclaimers

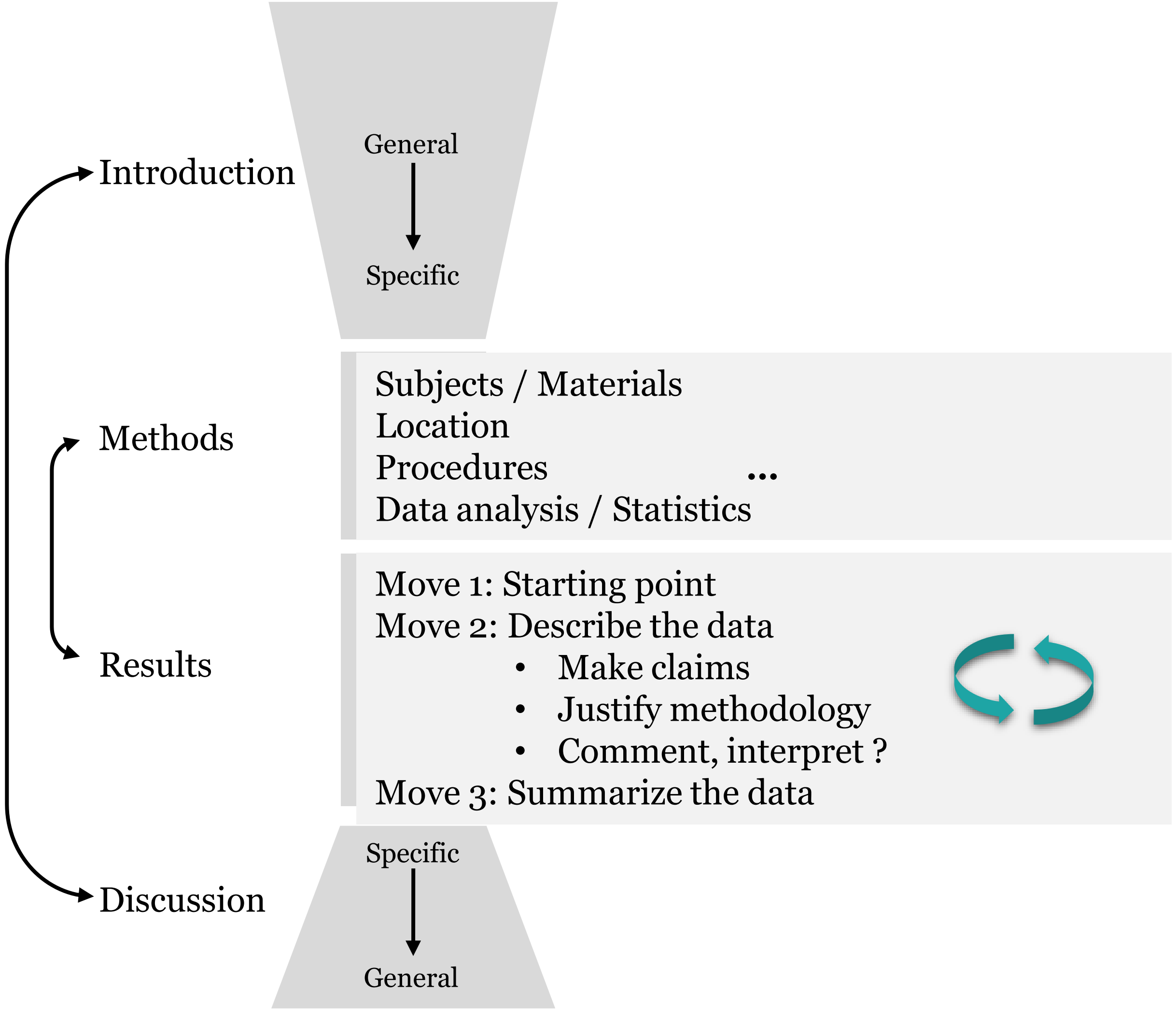
First person pronouns (I/we) mostly used

Research Article Layout

From Swales & Feak,

Academic Writing for Graduate Students, 3rd Ed.

The Methods and Results sections mirror each other to a large extent.



Task: Examine some articles/theses

- Is there a methods (and materials) section? If not, is there another section that describes how the study was conducted? What is it called?
- How long is it and what does it contain?
- Are there subheadings/subsections in the Methods section? What are they?

Other names for the Methods section

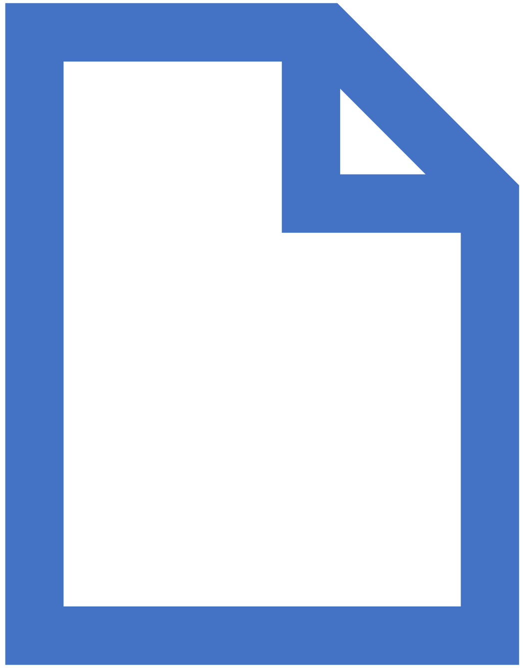
- Materials and Methods
- Patients and Methods
- Study Design
- Experimental section
- And other similar names

Goals of the Methods section

Allow readers to

1. understand how and why the experiments were performed,
2. better understand the remainder of the paper and how the results and conclusions derived from the experiments,
3. be able to **reproduce** the study with an expectation of success, and
4. acknowledge that the results and conclusions are valid based on the strength of the methods and study design,
5. **establish the credibility of your approach, thus giving readers a reason to believe your findings.**

Making sure to include the important details about *who*, *what*, *when*, *where*, *how*, and *why* in the study can help achieve these goals.



Goal of reproducability

https://www.nature.com/articles/d41586-023-03177-1

Reproducibility trial: 246 biologists get different results from same data sets

2016 *Nature* survey[3], :in the field of biology alone, over 70% of researchers were unable to reproduce the findings of other scientists and approximately 60% of researchers could not reproduce their own findings.

Six factors affecting reproducibility in life science research and how to handle them (nature.com)

- American Society for Cell Biology distinguishes between:

1. *Direct replication*, attemptsto reproduce a previously observed result by using the same experimental design and conditions as the original study;
2. *analytic replication*, which aims to reproduce a series of scientific findings through a reanalysis of the original data set;
3. *systemic replication*, which is an attempt to reproduce a published finding under different experimental conditions (e.g., in a different culture system or animal model); and
4. *conceptual replication*, where the validity of a phenomenon is evaluated using a different set of experimental conditions or methods.

Factors contributing to lack of reproducibility

1. *A lack of access to methodological details, raw data, and research materials*
2. *Use of misidentified, cross-contaminated, or over-passaged cell lines and microorganisms*
3. *Inability to manage complex datasets*
4. *Poor research practices and experimental design*
5. *Cognitive bias:* ways that judgement and decision-making are affected by the individual subjective social context that each person builds around them. They are errors made in cognitive processes that are due to personal beliefs or perceptions
6. *A competitive culture that rewards novel findings and undervalues negative results*

(Six factors affecting reproducibility in life science research and how to handle them (nature.com))

Recommended best practices to address lack of reproducibility

1. *Robust sharing of data, materials, software, and other tools.* All of the raw data that underlies any published conclusions should be readily available to fellow researchers and reviewers of the published article
2. *Use of authenticated biomaterials*
3. *Training researchers on statistical methods and study design*
4. *Pre-registration of scientific studies*
5. *Publish negative results*
6. *Thorough description of methods:* Researchers should clearly report key experimental parameters, such as whether experiments were blinded, which standards and instruments were used, how many replicates were made, how the results were interpreted, how the statistical analysis was performed, how the randomization was done, and what criteria were used to include or exclude any data.

- Six factors affecting reproducibility in life science research and how to handle them (nature.com)

Basic elements of a Methods section

- Study design
- Setting and subjects
- Data collection
- Data analysis
- Ethical approval
- Justification for choice of method or materials

Outline of the Methods section (See table with comprehensive list.)

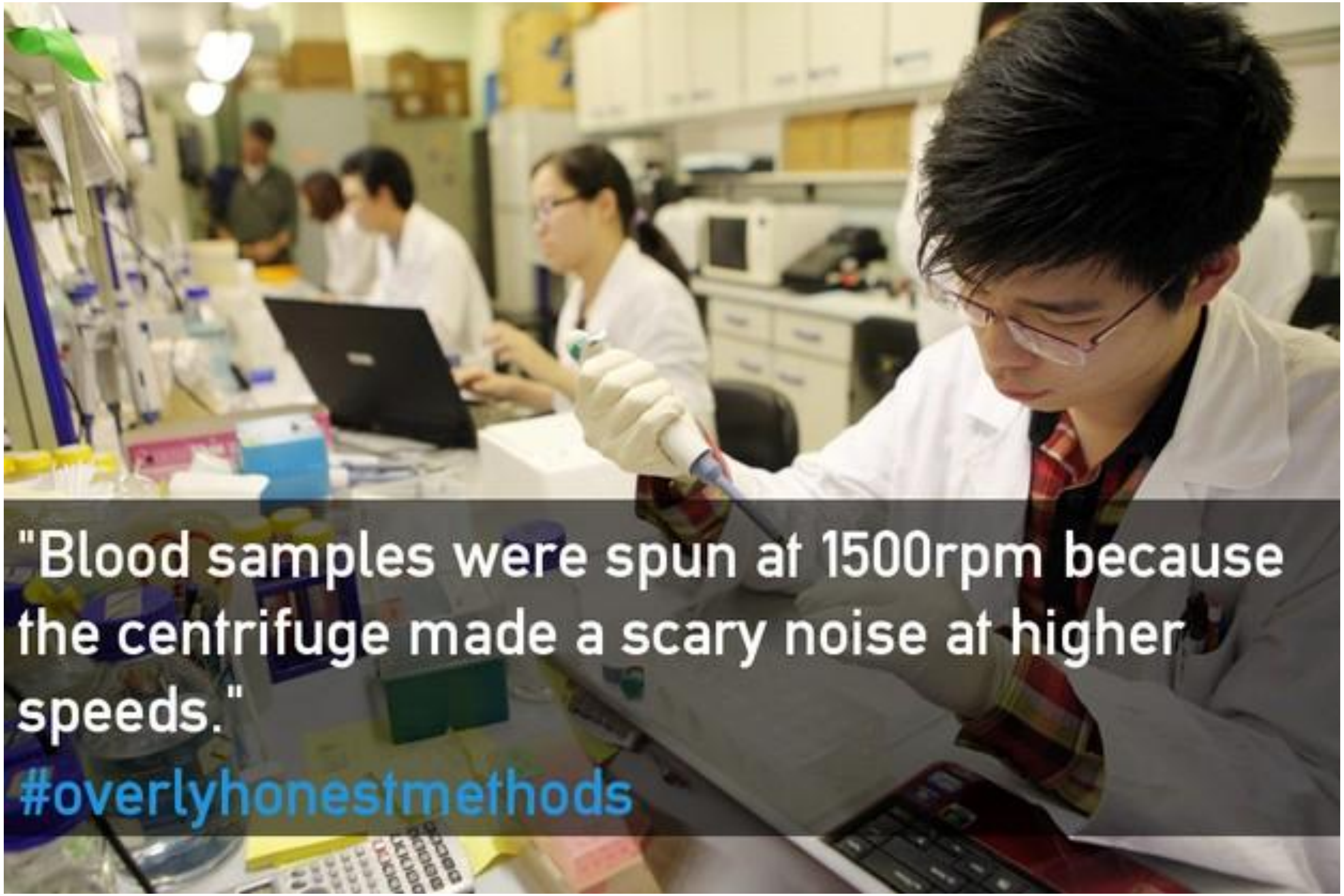
- The Methods section should be divided into subsections with associated subheadings. The use of subheadings helps organize the material in the reader's mind.
- What did you plan to do?
- How did you plan to do that?
- Why did you choose that specific approach/ method?
- What was your hypothesis? / What were your research questions?
- How did the method fit the questions?
- Which topics do you need to cover to explain your methods (e.g. subjects, treatment groups, etc)?

Table 1. Who, what, when, where, how, and why questions to consider when writing the Methods section (Annesley 2010:898)

Who : Who maintained the records? Who reviewed the data? Who collected the specimens? Who enrolled the study participants? Who supplied the reagents? Who did the statistical analyses? Who reviewed the protocol for ethics approval? Who provided the funding?
What : What reagents, methods, and instruments were used? What type of study was it? What were the inclusion and exclusion criteria for enrolling study participants? What protocol was followed? What treatments were given? What endpoints were measured? What data transformation was performed? What statistical software package was used? What…
When : When were specimens collected? When were the analyses performed? When was the study initiated? When was the study terminated? When were the diagnoses made?
Where : Where were the records kept? Where were the specimens analysed? Where were the study participants enrolled? Where was the study performed?
How : How were samples collected, processed, and stored? How many replicates were performed? How was the data reported? How were the study participants selected/recruited? How was the sample size determined? How were study participants assigned to groups? How were endpoints measured? How were control and disease groups defined?
Why : Why was a species chosen (mice vs rats)? Why was a selected analytical method chosen? Why was a selected experiment performed? Why were experiments done in a certain order?

'Why' question: justification - some common mistakes in the Methods section and recommendations to fix them (Busse and August 2020)

Pitfall	Recommendation
The author only describes methods for one study aim, or part of an aim	Make sure the methods are complete. Check that the methods describe all aspects of the study reported in the manuscript.
There is not enough (or any) justification for the methods used.	Justify each approach and variable. Justify your choice of methods because it greatly impacts the interpretation of results. State the methods you used and then defend those decisions. For example, justify why you chose to include the measurements, covariates, and statistical approaches.


"Blood samples were spun at 1500rpm because the centrifuge made a scary noise at higher speeds."
#overlyhonestmethods

Clinical sciences	Life sciences
• study design • setting and subjects • data collection • data analysis & statistics. Indicate tests and software used • ethical approval (patients)	• materials (e.g. animals, cell lines, antibodies) • experimental procedures • data analysis or statistical methods • ethical approval (animals)

- Gifts of materials: state this either in the methods or in the **acknowledgements** section

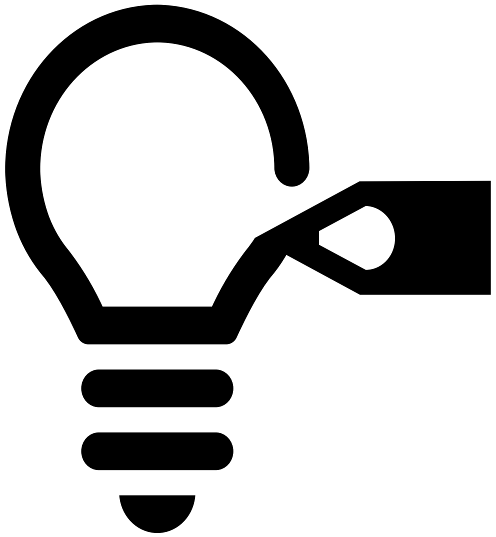
Some study design examples – covered by STPH courses

- Meta-analysis
- Systematic review
- Randomized controlled trial
- Observational studies
- Case reports & case series

7 Moves in the Methods section

1. **Overview**: a short summary of the research method(s), including justification if necessary
2. **Research aims, questions or hypotheses** (here and/or in Introduction)
3. **Subjects and/or materials**
4. **Location**: where the research took place
5. **Procedure**: process used to obtain the data
6. **Limitations**: shortcomings of the method & an explanation
7. **Data analysis** – how the data was analysed

Task: Moves in the Methods section



Consult some articles/theses in your field.

1. Can you **identify** some of the moves indicated above?
2. Is the level of detail included sufficient for the study to be replicated by others?
3. How is the information organized? Would it help the reader to include any subheadings?
4. What verb tense dominates?
5. Active voice/Passive voice use?
6. Justification given for the methods?
7. Limitations of methods expressed?

Qualitative research methods

- Description of the subjects and location might be quite elaborate.
- More explicit information given about the researcher, his/her past experience with the topic, his/her possible role as participant observer, the setting and the subjects.
- Indicate the various types of data collection used (interviews, observations, focus groups, document analysis, etc.)
- Is the type of language use the same? Objectivity?
- The same study can combine qualitative and quantitative methods.

Read the excerpt of a qualitative Methods section in the script as illustration of some of these qualities.


QuestionPro
ONE-ON-ONE INTERVIEW
CASE STUDY RESEARCH
FOCUS GROUPS
QUALITATIVE RESEARCH METHODS
RECORD KEEPING
ETHNOGRAPHIC RESEARCH
PROCESS OF OBSERVATION

Task: Compare qualitative and quantitative methods sections

1. What similarities are there?
2. What differences are there? Highlight them. Consider and describe a wide array of possible differences, e.g. language use, type of methods, replicability/reproducibility, ethical issues
3. What verb tense dominates?
4. Active voice/Passive voice use?
5. Justification given for the methods?
6. Limitations of methods expressed?

Quantitative research

Data Driven

Numbers & percentages

Concrete & objective

HOW MUCH? HOW MANY?

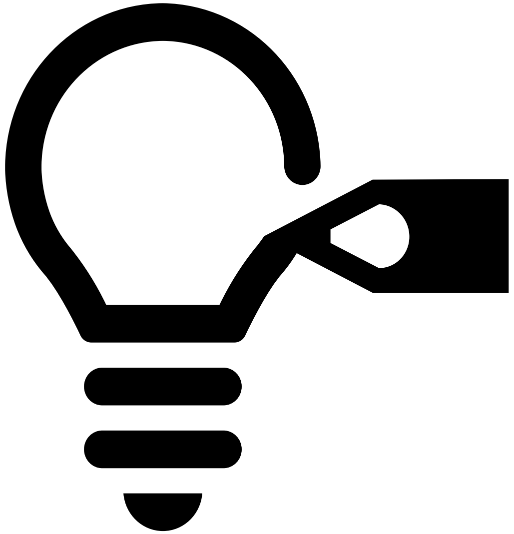
Qualitative research

Design Thinking

Quotes & expressions

Abstract & subjective

WHY? WHAT? HOW TO FIX?



Language focus task: Examine methods sections in some articles/theses

1. How does the first sentence begin? Is that a pattern that is followed? Highlight and describe patterns.
2. If there is there any time indication, highlight it.
3. Do the authors refer to themselves? How?
4. What tense is mainly used?
5. Is the passive voice used? Is the active voice used? How are they used?

Language use in Methods

Purpose statements , e.g.

- In an effort to evaluate…
- To/In order to establish…
- To determine the cost…

Time/Temporal phrases e.g.

- During the data collection…
- Prior to collecting this information…
- In the follow-up phase of the study, we…

Causal/connective phrases to justify the methodology

- Based on previous reports/feedback from the pilot study
- On the basis of the literature review…
- Because of privacy issues...

Focus on the added **bold**, *italics* and <u>underlining</u> to focus on the language features

To **detect groups among the specimens and extract the variables that best diagnose the groups**, principal components analysis (PCA) *was used*. <u>Before conducting the analysis</u>, all measurements *were standardized* so that each variable would have a mean of 0 and a standard deviation of 1. <u>For the PCA</u>, only continuous characters *were included*. **In order to avoid weighting characters**, *we excluded* characters that are probably genetically redundant, as revealed by high values for the Pearson correlation coefficient between all possible pairs of characters.

Some typical language features used in the Methods extract on the previous slide:

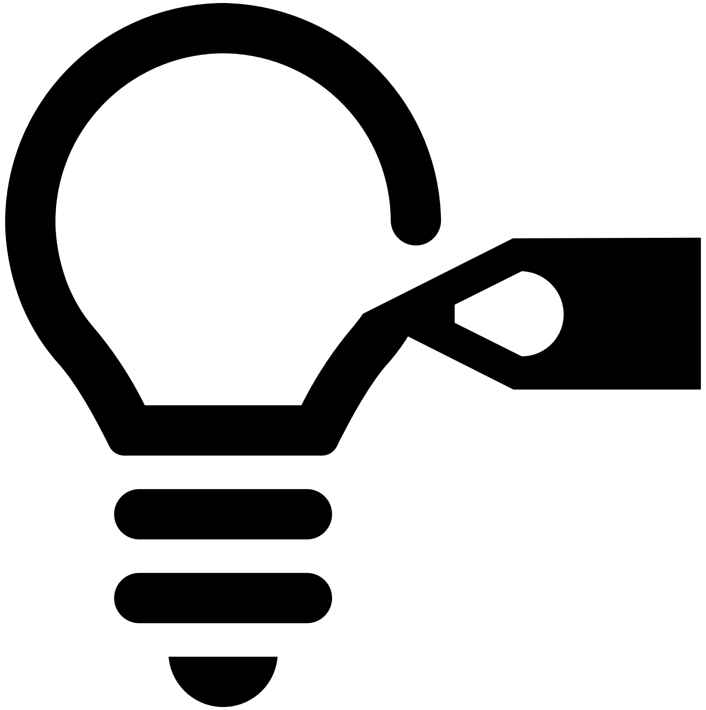
- The *we* form as well as the passive voice was used.
- There is a time indication <u>Before conducting the analysis</u>
- and a clarifying phrase <u>For the PCA</u> at the opening of the 3rd sentence.
- Use of the purpose statements T**o detect groups …** and **In order to avoid…** that explain and justify the chosen procedures.
- As PCA and the Pearson correlation coefficient are well-known methods, no explanation was given, but if the methods are less well known, one has to justify their use.

Why use transition / linking phrases?

- ➢ Avoid monotony
- ➢ Improve cohesion and text flow
- ➢ Introduce a new experiment
- ➢ Describe why the experiment was performed
- ➢ Introduce new paragraphs

Examples:

- *After* mixing for 1 minute, we added 7 mL methylene chloride ...
- *To evaluate* ion suppression, extracts were prepared ...
- *Because* the cells did not adhere to polypropylene, fibroblasts were grown ...
- *Because of* its signal-enhancing properties, cinnamic acid was added ...
- *Based on* previous reports of calcitrol-induced inhibition, we added calcitrol…



Task: Language use: linking /transition phrases

1. **Identify** and highlight some purpose statements, temporal phrases (time indications) and connective/causal phrases in articles.
2. Which ones are most frequent?

Methods: linguistic considerations

Avoid nominalisations: use an action verb rather than a weak verb + noun.

‘weak’ verb examples: to do, to perform, to carry out

Examples of nominalisations: analysis, classification, categorisation, selection

Action verbs: analyse, classify, categorise, select

Genotypes were aligned and analysed using the UCSC GenomeBrowser.

NOT: Alignment and analysis of genotypes were done using the UCSC GenomeBrowser.

Language use: Simple past tense, passive voice as well as active with ‘We’

The Methods section is mostly written in the Simple Past tense.

- **We collected** samples every morning for 21 days. (Active voice)
- **Samples were collected** every morning for 21 days. (Passive voice)

Use this tense refer to experiences you had or studies that were done at a definite time in the past**.** There is normally a time indication in the past: you can usually answer the question ‘when did this happen?’.

- He *participated* in a clinical trial in 2020.
- The results *were published* four months ago.
- At the beginning of the previous century, life expectancy *was* 20 years less than it is now.
- Do NOT use the present perfect tense (has/have + past participle) to refer to completed actions in the past. See next slide.

Language use: Present perfect tense (has/have + past participle) connects the past and the present

Use the **present perfect tense** to

- describe trends over a period of time in the (recent) past
- describe events that began in the past and are still continuing
- describe events that have recently happened and are relevant to the present
- refer to an **indefinite time** in the past.

- Some words that are commonly used with the present perfect tense are: *just, already, so far, up to now, since* and *for, ever* and *never*.
- Do NOT use this tense with a definite time indication in the past, e.g. ~~in 2018~~

PAST SIMPLE TENSE vs PRESENT PERFECT TENSE

Past Simple	Present perfect
Finished time. We lived in Japan in from 1995 to 1998.	Unfinished time. I have worked as a teacher since 2011.
Definite time. I saw the Eiffel tower in 2007.	Indefinite time. I have seen the Eiffel tower.
Series of finished actions. He read the book and then he watched the movie.	Experience. Have you seen this movie?
Repeated actions. He went to the cinema every weekend last year.	Result She has already watched this movie three times.

Present perfect tense in scientific writing

The present perfect tense (has/have + past participle form) is useful for referring to:

- earlier studies over a period of time up to the present
- generally accepted theories.

Frequently used in the Introduction section of an article, for example:

- Several recent studies *have demonstrated* that …
- Epidemiologists *have identified* the means of transmission of X…
-

Test yourself: Tenses

Do the exercises in the script to test yourself on the use of the present perfect and past simple tenses, as well as the passive voice.

Active voice

Tells us what a person or thing does.

The subject performs the action (verb) on the object.

Subject + verb + object

Example:

- Anna painted the house.
- The teacher always answers the students' questions.
- Ali posted the video online.

Passive voice

Tells us what is done to someone or something.

The subject is being acted upon.

Object + verb + subject

Example:

- The house was painted by Anna.
- The students' questions are answered by the teacher.
- The video was posted online by Ali.

Passive voice

Use the ACTIVE voice as far as possible, especially in the rest of the paper, as it is more readable.

Sometimes passive voice is necessary:

1. To emphasize the action rather than the actor.

- After filtration, the solution was poured into a …

2. To help text flow by keeping the subject and focus consistent throughout a passage.

- The research group recently presented a controversial **proposal** to conduct experiments in a nature reserve. After long debate, the **proposal** *was endorsed* by…

Do the exercises on the passive voice in the script.

Passive voice in scientific writing:

- Makes writing more impersonal and formal, but should not be overused:

- not reader-friendly (it is less easily readable)
- can make the writing appear wooden or static and hard to read
- slows the reader down if overused
- complex structure
- can be confusing and wordy – lack of clarity

- The next example shows that passive voice sentences are wordy – using the active voice is more concise:
-
  - *The passive voice almost always makes your message less clear.*

  - *Your message is almost always made less clear by using the passive voice.*

COHERENCE AND COHESION

Coherence is the logical unity and organization of ideas in a text.
Lack of coherence means that points within one paragraph are unrelated (or paragraphs within one text are unrelated).

Cohesion is the connection of sentences through linguistic means, which allow to achieve a smooth text flow effect. Cohesive elements include conjunctions, reference words, substitution and lexical devices such as repetition of words, collocations, and lexical groups.

What we need to know is that the text should not sound like a list of bullet points (lack of cohesion), and ideas should be placed in a logical order (coherence).

Cohesion and coherence

To make writing cohesive and coherent, one can repeat key words and synonyms, as well as *linking words* to guide the reader.

A cohesive text refers to other parts of the text (e.g. ‘the above-mentioned aims’) using words like *they, this, those, some* and *its* to create a well-organized piece of writing that will make it easier for the reader to follow the complex arguments in academic writing.

Family structure and society

Throughout history, a priority for every known society has been to develop a reliable arrangement for the procreation and rearing of children, an institution where members also cooperate to give each other emotional and economic support and where children receive their socialisation. This universal institution is the family. However, although the family performs a range of broadly similar functions in human societies, its form has varied widely according to the differing values and circumstances of those societies. (p. 271)

Summarizing nouns with this/these

- advance
- advice
- amount
- area
- argument
- change
- claim
- comment
- conclusion
- crisis
- criticism
- description
- development
- difficulty
- disagreement
- discussion
- drop
- estimate
- example
- explanation
- fall
- idea
- increase
- issue
- measure
- method
- number
- objective
- phenomenon
- policy
- Problem
- understanding
- proposal
- reduction
- remark
- report
- rise
- situation
- subject
- suggestion
- system
- trend
- view
- warning

Creating overall text (full-text) cohesion and coherence

- Introductory sentences in sections/ chapters
- Linking sentences between sections
- Use of linking devices such as conjunctions and careful repetition of key words
- Linking sections: link the Introduction and Conclusions sections (to show if you closed the 'gap' or solved the problem)
- Link the Methods to the Results (what did you find at each step)

Assignment 5: **Choice** of 5A: Methods **OR** 5B: Results . Same due date!

Assignment 5A:
Complete all three the following tasks:
1. Write down a title for your study in three different formats, as explored in this session. State which one you prefer.
2. Write a 200-300-word Methods section for your manuscript/article/proposal. Students who do not have methods sections yet, consider/envisage what their research methods might be and describe them or analyse methods sections in academic articles and write suitable phrases for this section. If your methodology is complicated and you need to use more words, you can do so.
Where applicable, highlight the Moves included: Overview, research aims, questions or hypotheses , subjects and/or materials, location, procedure, limitations, data
3.Write two acknowledgements that should altogether be no longer than 50 words.
OR
Assignment 5B:
Complete all three the following tasks:
1. Write down a title for your study in three different formats, as explored in this session. State which one you prefer.
2. Write a 200-300-word Results section for your manuscript. Students who do not have results yet, consider/envisage what results they expect or hope to get from the research, and describe and discuss them. You may, if you wish, add figures and figure legends for support (These do not count towards the word limit, nor do they have to be reviewed).
Incorporate effective linking devices (including summary nouns) to make your writing cohesive and coherent. These linking sentences as well as the requirements in the above-mentioned bullet points should be highlighted in your assignment.
3.Write two acknowledgements that should altogether be no longer than 50 words.
Reference all sources used in the text and in a short bibliography.
Put your assignment through the plagiarism check, Turnitin, on the ADAM workspace. Upload the Turnitin report too (not the receipt).